

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	175915	49.106034
1	[Git, Change, Create]	51835	14.469552
2	[Git, Change, Delete]	27780	7.754686
3	[Java, ByteCode, Member, Method]	13483	3.763731
4	[Java, ByteCode, Parameter]	13293	3.710693
5	[Git, Commit]	13078	3.650676
6	[File, Git]	10969	3.061957
7	[Git, Change, Rename]	10636	2.969001
8	[Java, ByteCode, Bound, ParameterizedType]	7279	2.031906
9	[Java, ByteCode, Bound]	7240	2.021020
10	[Java, ByteCode, Member, Field]	3606	1.006602
11	[Java, ByteCode, Bound, WildcardType]	2947	0.822644
12	[Java, Value, ByteCode, Annotation]	2931	0.818178
13	[Xml, Element]	2162	0.603514
14	[Java, ByteCode, Member, Constructor, Method]	2116	0.590674
15	[Xml, Text]	1450	0.404762
16	[Java, ByteCode, TypeVariable, Bound]	1111	0.310132
17	[Java, ByteCode, Member, Method, Lambda]	968	0.270214
18	[Type, File, Java, ByteCode, ResolvedDuplicate...	888	0.247882
19	[Type, File, Java, Class, ByteCode]	846	0.236158
20	[Json, Key]	702	0.195961
21	[Value, Json, Scalar]	685	0.191215
22	[Java, Value, ByteCode, Primitive]	672	0.187586
23	[Type, File, Java, ByteCode, JavaType]	649	0.181166
24	[Java, ByteCode, Member, Method, GenericDeclar...	578	0.161347
25	[Type, File, Java, ByteCode, ExternalType]	411	0.114729
26	[Git, Change, Copy]	334	0.093235
27	[Author, Git, Person]	299	0.083465
28	[Value, Array]	287	0.080115
29	[Committer, Git, Person]	252	0.070345

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

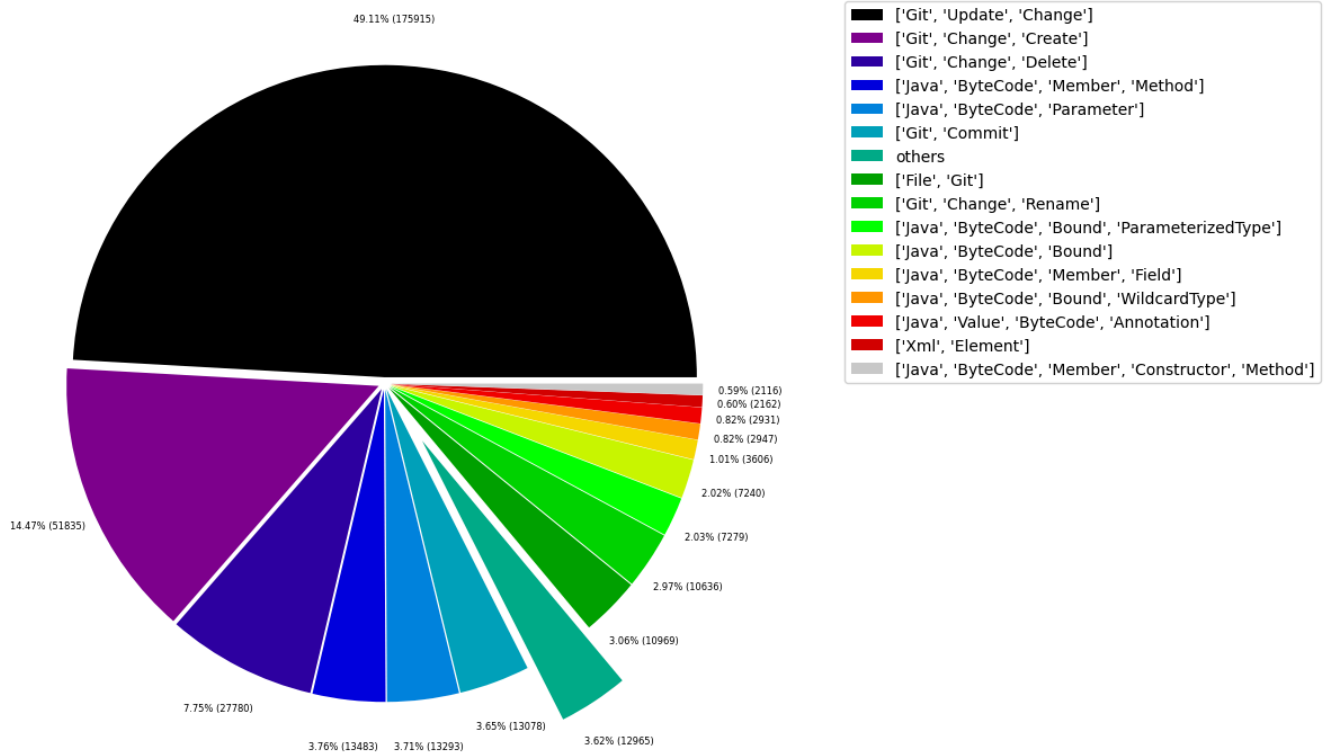


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000279
1	[Package, File, Json, NPM]	1	0.000279
2	[Repository, File, Git]	1	0.000279
3	[File, TS, Scan]	1	0.000279
4	[File, Json]	2	0.000558
5	[File]	3	0.000837
6	[Java, ByteCode, Member, Constructor, Method, ...]	4	0.001117
7	[Maven, Exclusion]	5	0.001396
8	[Value, Array, Json]	6	0.001675
9	[Dependency, NPM]	7	0.001954
10	[Type, File, Java, ByteCode, Void]	9	0.002512
11	[File, Maven, Xml, Pom, Document]	9	0.002512
12	[Java, ManifestSection]	9	0.002512
13	[File, Java, Manifest]	9	0.002512
14	[Artifact, File, Jar, Archive, Zip, Java]	9	0.002512
15	[File, Java, ServiceLoader]	10	0.002791
16	[File, Java, Properties]	12	0.003350
17	[Maven, ExecutionGoal]	16	0.004466
18	[Maven, PluginExecution]	16	0.004466
19	[Xml, Attribute]	18	0.005025
20	[jQAssistant, Rule, Concept]	19	0.005304
21	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.005862
22	[Maven, Configuration]	21	0.005862
23	[Maven, Plugin]	21	0.005862
24	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.006141
25	[Type, File, Java, ByteCode, Enum]	28	0.007816
26	[Type, File, Java, ByteCode, PrimitiveType]	30	0.008374
27	[Xml, Namespace]	36	0.010049
28	[Type, File, Java, ByteCode, Annotation]	44	0.012282
29	[Git, Branch]	45	0.012562

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

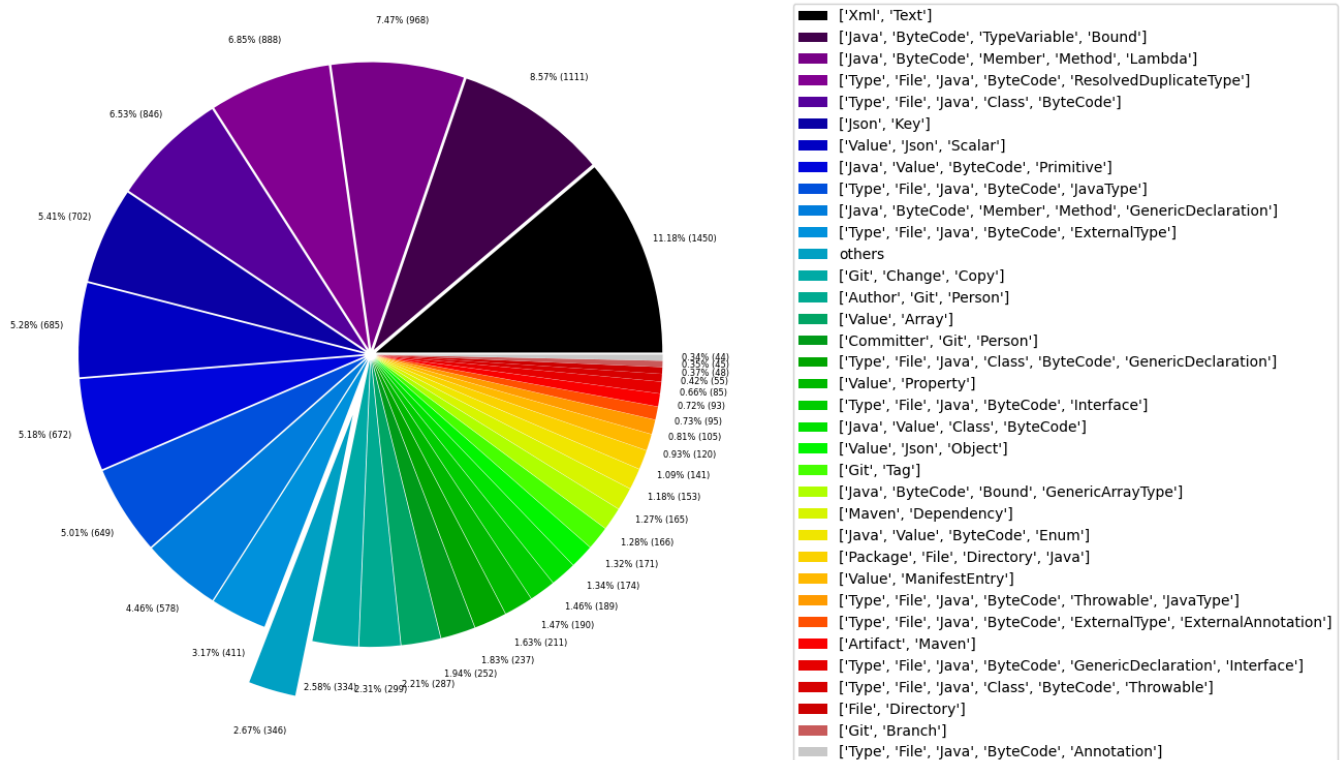


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

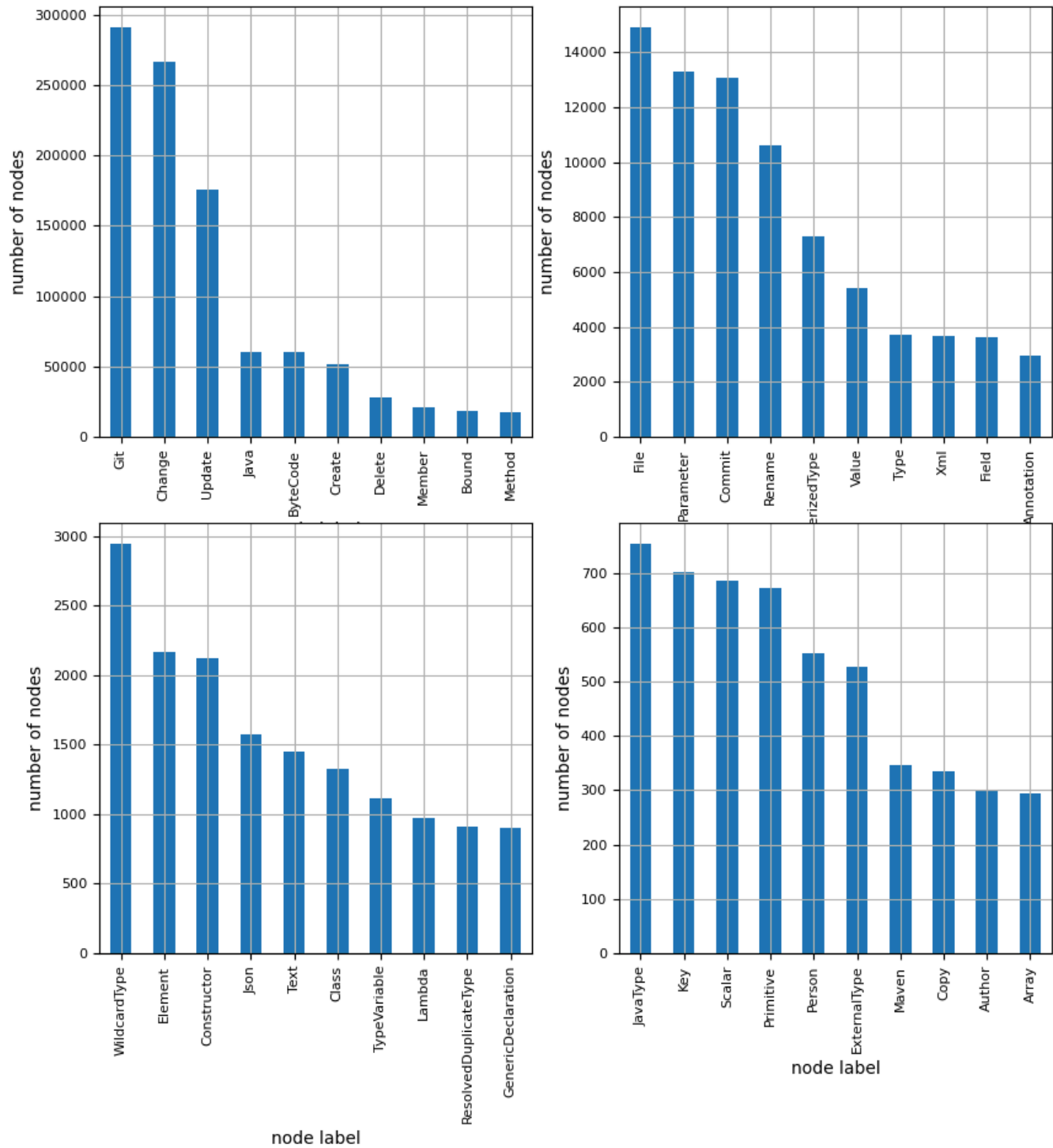
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	291315	81.319525
1	Change	266500	74.392508
2	Update	175915	49.106034
3	Java	60641	16.927715
4	ByteCode	60451	16.874677
5	Create	51835	14.469552
6	Delete	27780	7.754686
7	Member	20755	5.793683
8	Bound	18743	5.232040
9	Method	17149	4.787081
10	File	14930	4.167655
11	Parameter	13293	3.710693
12	Commit	13078	3.650676
13	Rename	10636	2.969001
14	ParameterizedType	7279	2.031906
15	Value	5428	1.515206
16	Type	3715	1.037029
17	Xml	3675	1.025863
18	Field	3606	1.006602
19	Annotation	2975	0.830460
20	WildcardType	2947	0.822644
21	Element	2162	0.603514
22	Constructor	2120	0.591790
23	Json	1570	0.438260
24	Text	1450	0.404762
25	Class	1327	0.370427
26	TypeVariable	1111	0.310132
27	Lambda	968	0.270214
28	ResolvedDuplicateType	910	0.254023
29	GenericDeclaration	904	0.252348
30	JavaType	754	0.210476
31	Key	702	0.195961
32	Scalar	685	0.191215
33	Primitive	672	0.187586
34	Person	551	0.153810
35	ExternalType	527	0.147110
36	Maven	346	0.096585
37	Copy	334	0.093235
38	Author	299	0.083465
39	Array	293	0.081790

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 1111671

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	266500	23.972920
1	MODIFIES	266500	23.972920
2	UPDATES	175915	15.824376
3	CREATES	62805	5.649603
4	DELETES	38416	3.455699
5	INVOKES	36759	3.306644
6	COMMITTED	26156	2.352854
7	DEPENDS_ON	22492	2.023260
8	OF_TYPE	21920	1.971806
9	DECLARES	21216	1.908478
10	OF_RAW_TYPE	17353	1.560983
11	HAS_PARENT	15829	1.423893
12	HAS	14429	1.297956
13	HAS_COMMIT	13078	1.176427
14	RETURNS	12844	1.155378
15	HAS_FILE	10969	0.986713
16	RENAMES	10636	0.956758
17	READS	9378	0.843595
18	HAS_ACTUAL_TYPE_ARGUMENT	8407	0.756249
19	HAS_NEW_NAME	6255	0.562666
20	OF_GENERIC_TYPE	5996	0.539368
21	RESOLVES_TO	5270	0.474061
22	SIMILAR	4078	0.366835
23	WRITES	3944	0.354781
24	CONTAINS	3936	0.354062
25	RETURNS_GENERIC	3593	0.323207
26	ANNOTATED_BY	2919	0.262578
27	REQUIRES	2230	0.200599
28	HAS_FIRST_CHILD	2162	0.194482
29	HAS_LAST_CHILD	2162	0.194482

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

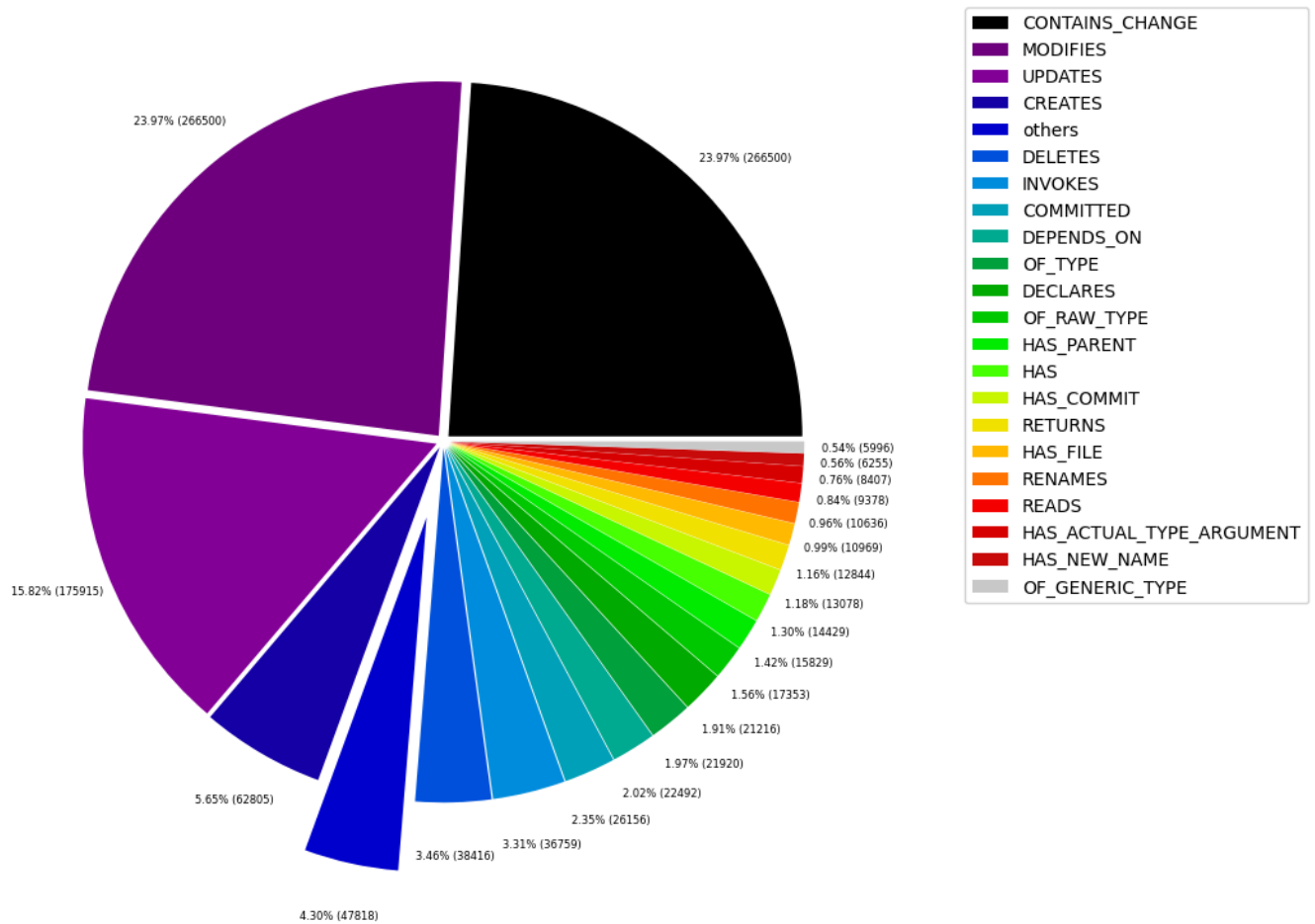


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

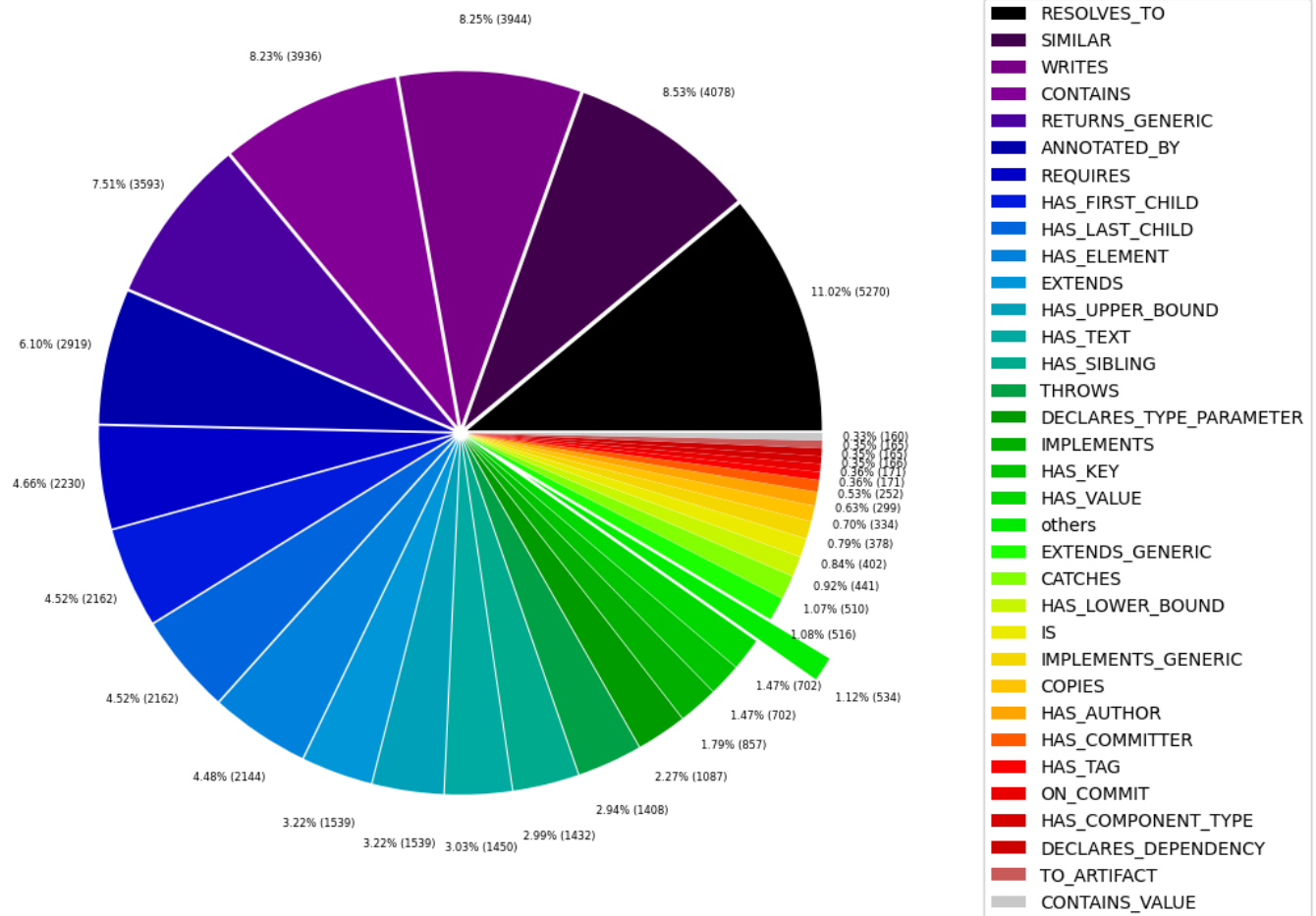
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000090
1	THROWS_GENERIC	5	0.000450
2	EXCLUDES	5	0.000450
3	DECLARES_DEV_DEPENDENCY	7	0.000630
4	DESCRIBES	9	0.000810
5	HAS_ROOT_ELEMENT	12	0.001079
6	HAS_GOAL	16	0.001439
7	HAS_EXECUTION	16	0.001439
8	OF_NAMESPACE	18	0.001619
9	HAS_ATTRIBUTE	18	0.001619
10	INCLUDES_CONCEPT	19	0.001709
11	IS_ARTIFACT	21	0.001889
12	HAS_CONFIGURATION	21	0.001889
13	USES_PLUGIN	21	0.001889
14	REQUIRES_TYPE_PARAMETER	24	0.002159
15	REQUIRES_CONCEPT	28	0.002519
16	DECLARES_NAMESPACE	36	0.003238
17	HAS_DEFAULT	39	0.003508
18	HAS_BRANCH	45	0.004048
19	HAS_HEAD	46	0.004138
20	COPY_OF	127	0.011424
21	CONTAINS_VALUE	160	0.014393
22	DECLARES_DEPENDENCY	165	0.014843
23	TO_ARTIFACT	165	0.014843
24	HAS_COMPONENT_TYPE	166	0.014932
25	ON_COMMIT	171	0.015382
26	HAS_TAG	171	0.015382
27	HAS_COMMITTER	252	0.022669
28	HAS_AUTHOR	299	0.026896
29	COPIES	334	0.030045

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Update, Change]	MODIFIES	[File, Git]	175915	175915	
1	[Git, Update, Change]	UPDATES	[File, Git]	175915	175915	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	175915	13078	
3	[Git, Change, Create]	CREATES	[File, Git]	51835	51835	
4	[Git, Change, Create]	MODIFIES	[File, Git]	51835	51835	
5	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	51835	13078	
6	[Git, Change, Delete]	DELETES	[File, Git]	27780	27780	
7	[Git, Change, Delete]	MODIFIES	[File, Git]	27780	27780	
8	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	27780	13078	
9	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	22506	13388	
10	[Git, Commit]	HAS_PARENT	[Git, Commit]	15820	13078	
11	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	13078	1	
12	[Author, Git, Person]	COMMITTED	[Git, Commit]	13078	299	
13	[Committer, Git, Person]	COMMITTED	[Git, Commit]	13078	252	
14	[Repository, File, Git]	HAS_FILE	[File, Git]	10969	1	
15	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	10636	13078	
16	[Git, Change, Rename]	DELETES	[File, Git]	10636	10636	
17	[Git, Change, Rename]	CREATES	[File, Git]	10636	10636	
18	[Git, Change, Rename]	MODIFIES	[File, Git]	10636	10636	
19	[Git, Change, Rename]	RENAMES	[File, Git]	10636	10636	
20	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	8503	13388	
21	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	8430	13388	
22	[File, Git]	HAS_NEW_NAME	[File, Git]	6255	10969	
23	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	6156	13293	
24	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3465	7240	
25	[Java, ByteCode, Bound, ParameterizedType]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3110	7279	
26	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2947	7279	
27	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, Bound, ParameterizedType]	2695	13293	
28	[Java, ByteCode, Member, Constructor, Method]	WRITES	[Java, ByteCode, Member, Field]	2519	2116	
29	[Java, ByteCode, Bound, ParameterizedType]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, TypeVariable, Bound]	2474	7279	

Graph Density

```
total_number_of_nodes (vertices): 358235
total_number_of_relationships (edges): 1111671
-> total directed graph density: 8.662464331363892e-06
-> total directed graph density in percent: 0.0008662464331363892
```